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Exploring crosslinguistic and genre variation in syntactic complexity: insights from a multilingual parallel corpus

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Abstract: Syntactic complexity varies not only between languages but also across textual genres, reflecting structural, stylistic, and communicative factors. This study explores the interplay of these two dimensions by analyzing six syntactic complexity metrics (SCMs) across 17 languages and four genres in a large multilingual corpus. We examine how SCMs correlate across languages and genres and assess which metrics best predict language versus genre. Correlation analysis reveals two robust metric clusters – clausal and phrasal. Inter-metric correlations are high in legal texts and movie subtitles, but substantially lower in parliamentary proceedings and fiction, likely due to greater internal variation in syntactic complexity within these genres. Using random forest classification, we show that SCMs strongly predict genre, with noun phrase metrics ranking highest. In contrast, mean dependency distance (mdd) and its standard deviation are most predictive of language. Misclassification tendencies confirm the affinity of English with Romance languages, previously observed in lexical studies. By emphasizing not only central tendencies (mean values) but also the dispersion of metrics within texts, this study adds methodological nuance to the measurement of syntactic complexity. It also confirms the need to account for both language and genre in crosslinguistic syntactic complexity research.

Keywords: syntactic complexity, genre variation, crosslinguistic variation, multilingual corpus, Universal Dependencies

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1 Introduction

The syntactic complexity of texts varies across languages, largely because of structural differences. It also differs significantly across textual genres (e.g., legal texts, fiction, or movie subtitles), reflecting stylistic preferences (Biber and Gray 2016: 132) and the functional and communicative needs of language users (Biber et al. 2024: 351; Ehret 2021: 386, 402; Xu and Li 2021: 215). Some crosslinguistic studies on syntactic complexity have acknowledged that their results may be biased by genre composition differences in corpora (Berdicevskis et al. 2018: 15; Brunato and Venturi 2023: 61; Sadeniemi et al. 2008: 195). However, no large-scale study has yet analyzed the joint impact of language and genre on syntactic complexity, mostly due to the lack of corpora covering both dimensions.

This study addresses that gap by exploring how syntactic complexity metrics (SCMs) correlate across languages and genres, and which metrics are more predictive of language-related variation vs. genre-related variation. We draw on InterCorp (Rosen et al. 2024; https://wiki.korpus.cz/doku.php/en:cnk:intercorp:verze16ud), a large, freely accessible multilingual parallel corpus that encompasses a variety of textual genres. The corpus includes Universal Dependencies (de Marneffe et al. 2021; https://universaldependencies.org) annotation which enabled us to enrich it with six SCMs at both sentence and text levels. Focusing on 17 languages and four genres, we address two research questions:

1. How do syntactic complexity metrics correlate with each other, and how do these correlation patterns differ across languages and genres?
2. Which metrics are more predictive of genre, and which are more predictive of language?

The paper is structured as follows. Section 2 defines our approach to syntactic complexity and outlines potential influencing factors. Section 3 describes the data and methodology. Section 4, the core of this study, presents and discusses the results with respect to the two central research questions. Section 5 concludes with a brief overview of the theoretical and methodological implications.

2 Framing syntactic complexity

Syntactic complexity can be defined in terms of general system complexity as the number and variety of elements and the elaborateness of their inter-relational structure (Rescher 1998: 1). Recent literature distinguishes two main perspectives on syntactic complexity (Biber et al. 2024: 348; De Clercq 2016: 3; Ehret 2021: 385; Miestamo 2009: 81): RELATIVE SUBJECTIVE COMPLEXITY, which concerns perceived processing difficulty or readability (Gruszczyński and Ogrodniczuk 2015), and ABSOLUTE (OBJECTIVE) COMPLEXITY, which describes inherent linguistic features. The latter approach either targets properties of the language system (GRAMMATICAL COMPLEXITY; Kortmann and Szmrecsanyi 2012: 9; Miestamo 2009) or the actual use of syntactic complexity features in texts (USAGE-BASED TEXT COMPLEXITY; Biber et al. 2024: 349). This paper adopts the second line, focusing on the absolute (structural) complexity of texts from a usage-based perspective.

Usage-based text complexity is grounded in the analysis of specific syntactic features of sentences in texts. It focuses on core indicators of complexity, namely the number, variety, and hierarchical organization of linguistic items. This approach has attracted much attention in recent years, particularly in computational linguistics (Berdicevskis et al. 2018; Brunato et al. 2020, 2022), translation studies (Xu and Li 2021), and register variation (Biber et al. 2024). The growth of the field has been fueled by the emergence of treebanks annotated in Universal Dependencies, which have expanded the possibilities for analyzing and searching for SCMs.

Findings from this usage-based line of research can be generalized to the level of registers or to the language as a whole (Berdicevskis et al. 2018: 10; Biber et al. 2024: 358). Nevertheless, the basic unit of analysis remains the individual texts and sentences. Importantly, the syntactic complexity of sentences and texts is shaped not only by structural properties of the language – for example, the inventory of available items – but also by usage-related factors, including the functional and communicative needs of language users and stylistic traditions associated with specific textual genres and registers.[1]

Structural differences between languages involving differences in syntactic complexity of sentences have been widely studied in contrastive linguistics. These studies, mostly limited to the comparison of two or three languages within one textual genre, have identified specific linguistic features underlying the differences, such as the use of (relative) subordinate clauses in French where English uses coordinate clauses (Cosme 2006). Another key factor is the degree of syntactic compression (Lehmann 1988), which allows for more concise and hierarchically organized sentence structures. There has been considerable interest in crosslinguistic analysis of converbs (for definition, see Coupe 2006). Studies show shifts in syntactic complexity when languages that frequently use these forms (*gérondif* and *participe présent* in French, *deeprichastie* in Russian, adverbial participles in *-ing* in English, etc.) are translated into languages that

1 In translations, syntactic complexity may be influenced by the source text and source language (i.e., the language from which the text is translated). However, research in translation studies shows that modern translators' awareness of this risk can paradoxically lead to normalization, i.e., the exaggeration of structural features typical of the target language (seminal studies include Toury 1995 and Baker 1996; for a recent overview of such research, see, e.g., Lefer and Vogeleer 2013 and Lefer and Granger 2022).

prefer finite verbs in such contexts, typically in coordinate clauses or VP-conjuncts, such as Czech or Norwegian (for Russian, Norwegian, and English, see Filiouchkina Krave 2012; for Czech, French, and Polish, see Nádvorníková 2024). Importantly, most crosslinguistic differences in syntactic complexity found in these studies do not stem from the absence of a structural element (e.g., a converb). Instead, they reflect genre-related and style-related preferences for particular strategies of information encoding and clause linking.

In register variation studies, two major registers have been identified with respect to syntactic complexity: spoken ("involved") registers, characterized by clausal complexity, and written ("informational") registers, characterized by phrasal complexity, especially elaborated NPs (Biber et al. 2024: 350). Biber and Gray (2016: 305) consider this distinction as potentially universal. However, a large register variation study on Czech showed that the phrasal–clausal (nominal–verbal) distinction is clearly manifested within written genres in this language (Cvrček et al. 2021: 356): nominal constructions occur in highly static informational texts (administrative or natural sciences), whereas (finite) verbal constructions are typical in fiction. These findings suggest that both language and register (or genre) variation can be decisive in shaping the syntactic complexity of sentences and texts.

3 Data and methods

The analyses are based on a subset of the InterCorp multilingual corpus (Rosen 2023). Its current release (release 16ud; https://wiki.korpus.cz/doku.php/en:cnk:intercorp:verze16ud) contains 5.3 billion words in 62 languages, including genres such as fiction, subtitles, legal documents, parliamentary proceedings, journalism, and 18 Bible translations. For 48 languages, InterCorp is annotated according to the Universal Dependencies standard, which enabled its enrichment with six SCMs at the sentence and text levels (see Section 3.2), making this study possible.[2]

For this study, we compiled a subcorpus of InterCorp by selecting parts that balance maximizing language and genre diversity with ensuring a sufficient number of texts for each language–genre combination. In what follows, we first present this subcorpus (Section 3.1), then describe the SCMs used in the analysis (Section 3.2), outline the data aggregation and sampling procedures (Section 3.3), and finally introduce the methods of data analysis examining relationships among the metrics and identifying those that best distinguish languages and genres (Section 3.4).

2 See also a table with average SCM values by language and genre:
https://wiki.korpus.cz/doku.php/en:cnk:intercorp:verze16ud#detailed_statistics.

3.1 The subcorpus

Our subcorpus covers 17 languages from six different genera: Bulgarian, Czech, Polish, Slovak, Slovene (Slavic); Danish, German, English, Dutch, Swedish (Germanic); French, Italian, Spanish, Portuguese (Romance); Finnish (Finnic); Hungarian (Ugric); and Latvian (Baltic). Each language is represented across four (textual) genres: fiction, subtitles (Open Subtitles; http://www.opensubtitles.org/), legal texts (Acquis Communautaire; https://ec.europa.eu/jrc/en/language-technologies/jrc-acquis), and parliamentary proceedings (Europarl; http://www.statmt.org/europarl/). The resulting predominance of Indo-European languages reflects the requirement to ensure a sufficient number of texts per language and genre combination, which excludes languages not available from EU sources. The size of the subcorpus and of the specific genres are given in Table 1; in total there are more than 3 billion words.[3]

Genre	Words	Sentences	Texts
Fiction	368,439 K	29,067 K	5 K
Acquis	347,282 K	23,209 K	308 K
Europarl	229,637 K	11,286 K	1,121 K
Subtitles	2,401,417 K	468,549 K	582 K
Total	3,346,776 K	532,111 K	2,016 K

Tab. 1: The size of the subcorpus for the four genres (in thousands).

For the reliability of the crosslinguistic analysis, it is relevant that each genre in Table 1 includes both original and translated texts, with the latter possibly reflecting translation-specific features, such as interference from the source language. However, in our data, such an effect should be minimized by the fact that these texts are mostly products of professional translators (Fiction, Acquis, and Europarl). The main exception is the Subtitles section of the subcorpus. But even in this case, a recent study (Levshina 2017b) found that these texts do not differ substantially from non-translated ones.

3.2 Selected metrics of syntactic complexity

As discussed in Section 2, syntactic complexity can be approached from multiple perspectives and is regarded as a multifaceted (Brunato and Venturi 2023: 61) or multi-

3 For more details, see the subcorpus statistics at https://jakobson.korpus.cz/~rosen/INTERCORP/VanguardSample/SubcorpusSizes.html.

dimensional (Biber et al. 2024: 351) phenomenon. A wide range of metrics has been discussed and summarized in the literature (see Jagaiah et al. 2020).

The SCMs chosen for this study had to capture different aspects of syntactic complexity: the depth and width of the syntactic tree and the variation of the syntactic unit category. Selecting suitable SCMs for a large multilingual corpus also required meeting several additional criteria: (i) feasibility within the constraints of annotation, storage, and available tools; (ii) reliability/comparability across languages; and (iii) transparency for potential reuse in future research. The resulting set of six metrics represents a balanced compromise (Table 2): it is diverse enough to reflect multiple dimensions of syntactic complexity while remaining practical and consistent for application at both the sentence and text levels (via mean values).

Category → **Dimension ↓**	**Word**	**Noun phrase**	**Clause**
Width	sLength	maxNPLength	subRatio
Depth	mdd	maxNPDepth	maxTreeDepth

Tab. 2: The six selected SCMs organized by targeted tree dimension and syntactic category. Two of the metrics (mdd and subRatio) extend across more than one dimension.

The syntactic tree in Figure 1 illustrates how these metrics are applied in practice. The example sentence and the whole corresponding text are annotated with the six SCM values, shown in Table 3.

Sentence length (sLength), measured in words, is a standard, easily interpretable metric (Liu et al. 2024; Szmrecsanyi 2004). Like in other measures, punctuation is ignored.

Mean dependency distance (mdd) is defined as the average number of word boundaries between syntactic heads and dependents (Liu 2008). The dependents in Figure 1 are relatively close to their heads, with at most one word in between, which results in the relatively low value of 1.89. Unlike sLength, mdd reflects syntactic structure and its interaction with word order.

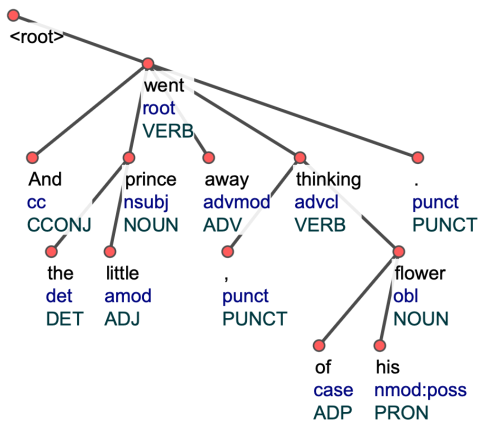

Fig. 1: Syntactic tree for the sentence *And the little prince went away, thinking of his flower* (Antoine de Saint-Exupéry, *The Little Prince*, translated by Katherine Woods), parsed with UDPipe (https://lindat.mff.cuni.cz/services/udpipe/, model english-ewt-ud-2.15-241121). InterCorp shows the same tree.

Metric	Sentence	Text
sLength	10	10.25
mdd	1.89	2.36
maxNPLength	3	3.84
maxNPDepth	1	1.34
subRatio	2	1.82
maxTreeDepth	1	0.69

Tab. 3: SCMs for the sentence in Figure 1 and their average values for the whole text, according to InterCorp (release 16ud).

The following two metrics focus on noun phrase complexity. The metric maxNPLength gives the number of words in the longest noun phrase in a sentence. In Figure 1, *the little prince* is responsible for its value of 3. On the other hand, maxNPDepth captures structural complexity of noun phrases by counting the number of dependent nodes in the longest dependency chain within any NP in the sentence. In Figure 1, there are two NPs, both with just a single level of dependents, thus maxNPDepth is 1. Like in the other depth-related metrics except mdd (i.e., subRatio and maxTreeDepth), function words are ignored.

The remaining two metrics target clauses. The metric maxTreeDepth measures the depth of the longest chain of embedded clauses in a sentence, defined as the number of embeddings leading to the most deeply nested clause. The clauses can also be nonfinite,[4] so the form *thinking* in Figure 1 introduces an additional embedding, resulting in a maxTreeDepth of 1 rather than 0. Subordination ratio (subRatio) is defined, following Jagaiah et al. (2020: 2600, 2622), as the ratio of T-units (T) and subordinate clauses (S) to T-units, measuring the contribution of subordination to syntactic complexity (see Equation 1; cf. also Beaman 1984).[5]

$$subRatio = \frac{T + S}{T} \tag{1}$$

Unlike maxTreeDepth, which focuses on the deepest syntactic sub-tree, subRatio measures complexity due to clausal embeddings for the whole sentence. Since there is just a single T-unit in Figure 1 (i.e., no coordination of main clauses), its value is 2.

3.3 Data aggregation and sampling

Each sentence in the subcorpus is associated with its SCMs and metadata (language, genre, text ID). For more efficient crosslinguistic and cross-genre comparison, the basic unit of analysis is the text. Sentence-level SCMs are aggregated as mean and standard deviation per text. To ensure size balance across languages and genres, samples are created based on the smallest group, with other groups populated randomly. As a result, the sampled texts compared across languages are not parallel versions of the same text.

3.4 Data analysis

Pearson correlation coefficients (r) were calculated for each pair of metrics, separately by genre and language. The results (see Section 4.1) were visualized using hierarchical clustering and heatmaps, based on naive (unweighted) averages across categories to avoid bias from unequal group sizes.

To assess which metrics best distinguish between languages and genres, we applied random forest classification models (see Section 4.2). This method enabled us to identify the most informative features – that is, the metrics and their within-text standard deviations – via feature importance analysis. Models were trained on 70% of the texts, with the remaining 30% used to evaluate accuracy and generate confusion matrices.

4 Note that clausal complexity in Biber's terms (see Section 2) focuses on finite clauses. However, reliably distinguishing finiteness in a crosslinguistic way using UD annotation is challenging, and thus was not included as a feature in the SCMs chosen for InterCorp.

5 T-unit is "one main clause with all subordinate clauses attached to it" (Hunt 1965: 20).

Two classifier sets were trained: one for genres and one for languages. For each, we saved the trained models, confusion matrices, accuracy and F1 scores, and feature importance comparisons. Additional visualizations (https://doi.org/10.17605/OSF.IO/KWXVS) include selected decision trees and plots showing text positions relative to the most important features.

4 Results

The annotated, aggregated, and sampled data offer various options for exploring relationships among the SCMs, languages, and genres represented in the subcorpus. In Section 4.1 we investigate patterns of pairwise correlations between the metrics, aiming at generalizations concerning the emergence of such patterns given specific languages and/or genres. The findings lead us to Section 4.2, where we test how the metrics distinguish between various languages and various genres and which metrics are most successful in this task.

4.1 Correlation of metrics across languages and genres

For each genre and language, correlation analysis was conducted on text-level metric values, derived by averaging the corresponding sentence-level scores. The hierarchical clustering tree in Figure 2 introduces an overview of tendencies in correlation between the six SCMs, across all languages and genres. The vertical axis corresponds to the distance between each cluster using the Ward method.

Starting from the bottom of the hierarchy, the two clausal metrics – that is, maxTreeDepth and subRatio – are the first merged pair, clustered together at a distance well below 0.1. This suggests a strong relationship between the two metrics across genres and languages. The two phrasal (NP) metrics, maxNPDepth and maxNPLength, are clustered higher, but still at a distance lower than 0.2, which also suggests a strong correlation: the longer the NP, the deeper its hierarchical structure (in terms of the level of the most deeply embedded dependent nodes; cf. the tree_depth metric in Brunato and Venturi 2023: 66).

The two remaining metrics, sLength and mdd, cluster together at the highest distance, yet still share properties, notably their dependence on sentence word count. This property likely also explains their correlation with the two phrasal (NP) metrics, even though these cluster at a higher distance. In contrast, the two clausal metrics (subRatio and maxTreeDepth), which capture hierarchical organization rather than element count, are the most specific. These results underscore the importance of considering both width

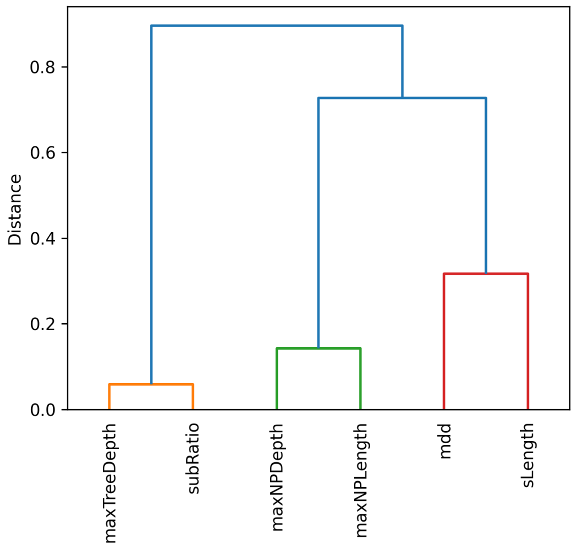

Fig. 2: Hierarchical clustering of syntactic complexity metrics based on correlation patterns.

and depth, as well as the distinctions between clausal and phrasal constituents (see Section 2).

Figure 3, zooming in on more specific data concerning the pairwise correlations between metrics across languages, shows a more refined picture. The dark columns confirm the high correlations of subRatio with maxTreeDepth (*r* exceeding 0.90 for almost all languages)[6] and of maxNPLength with maxNPDepth, with *r* ranging between 0.83 (Hungarian) and 0.87 (Slovenian).[7] The following ranks are dominated by correlations between various metrics and sLength, suggesting that all contribute to it. Notably, however, mdd shows a strong correlation only with sLength. Otherwise, correlations involving mdd are consistently the lowest, suggesting that mdd is our most specific and discriminating metric.

These observations on crosslinguistic differences and similarities regarding the correlation between SCMs need to be refined with respect to the specific textual genres represented in each language in the subcorpus. Figures 4 and 5 show the results with data split not only by language but also by genre (255 combinations in total, based on 15 metric pairs across 17 languages). While the overall pattern remains consistent across

6 The only exceptions are Dutch ($r = 0.89$) and Latvian ($r = 0.87$).

7 While the actual values of pairwise metric correlations – and thus the ranking of languages – may be affected by sample composition, the consistent *crosslinguistic* patterns suggest that the findings are robust.

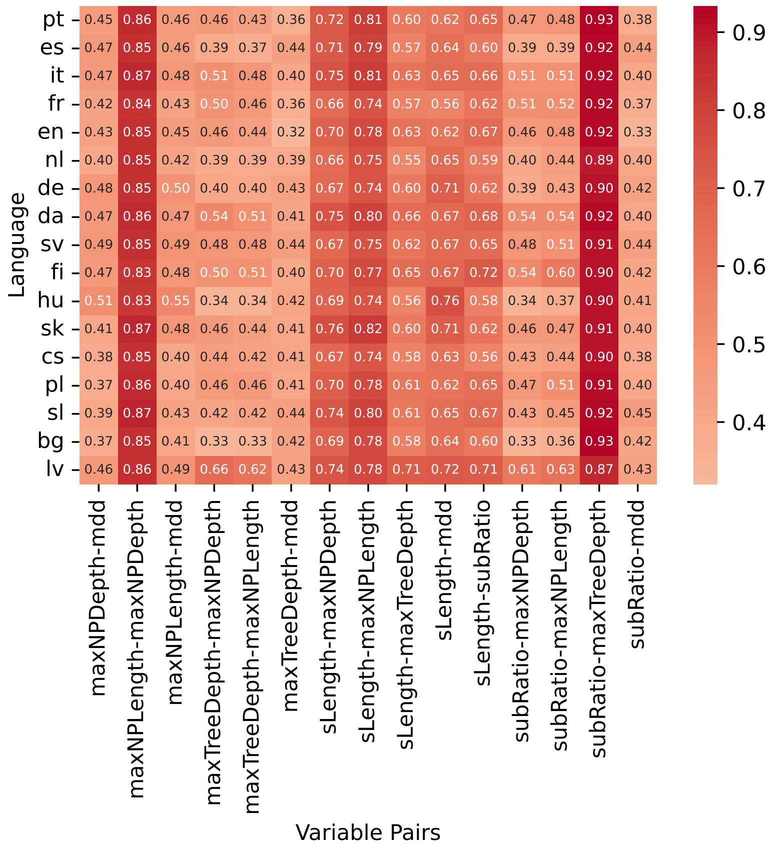

Fig. 3: Correlation patterns of syntactic complexity metrics across languages.

languages, particularly the high correlations between the two clausal metrics on one hand and the two NP metrics on the other, certain patterns are genre-specific, resulting in notable intra-linguistic variation in correlation values. Here, Acquis stands out: in 218 of the metric–language combinations (85.56% of all cases), it shows the highest correlation among the four genres and it is never the lowest. This suggests greater homogeneity in syntactic complexity, likely due not only to the highly constrained nature of legal discourse but also to the specific conditions of text production. In particular, Acquis texts are translated only from French, English, or both, which may introduce source-language interference.

Acquis is outperformed in metric-pair correlations only by Subtitles and only in two cases, which also show the strongest correlations in the corpus: the two clausal

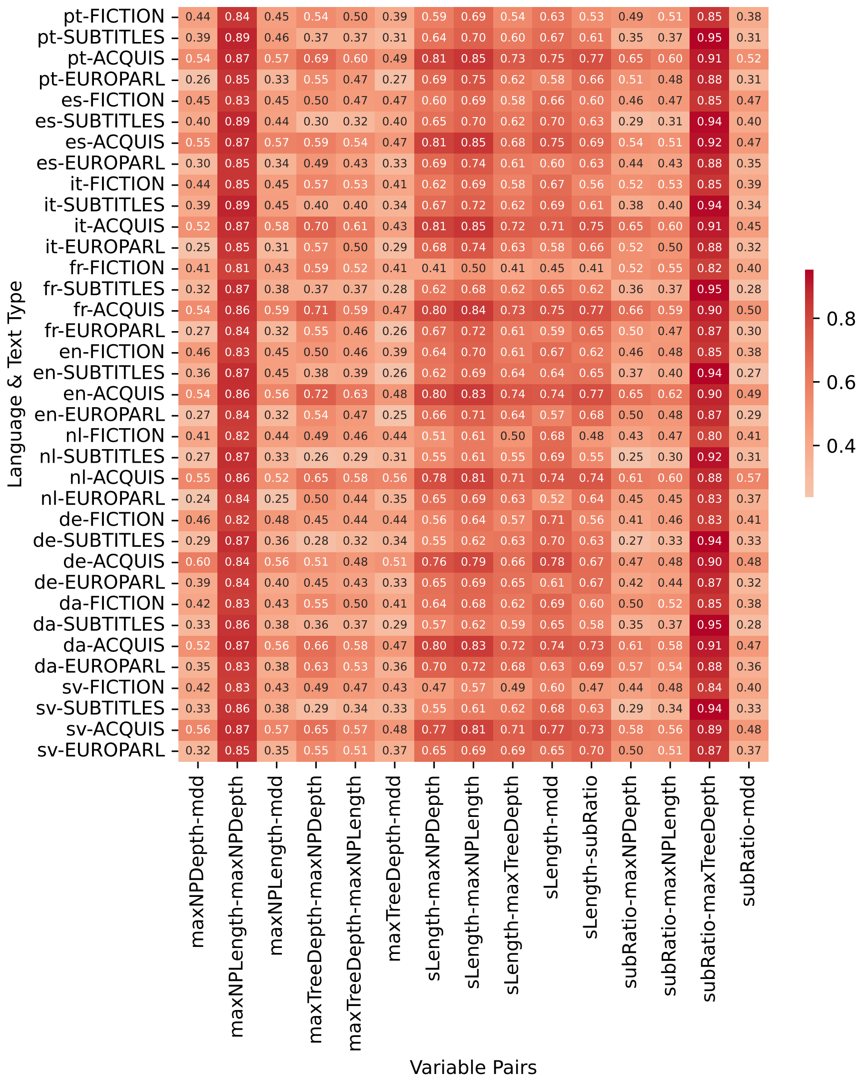

Fig. 4: Correlation patterns of syntactic complexity metrics across languages and genres (part 1).

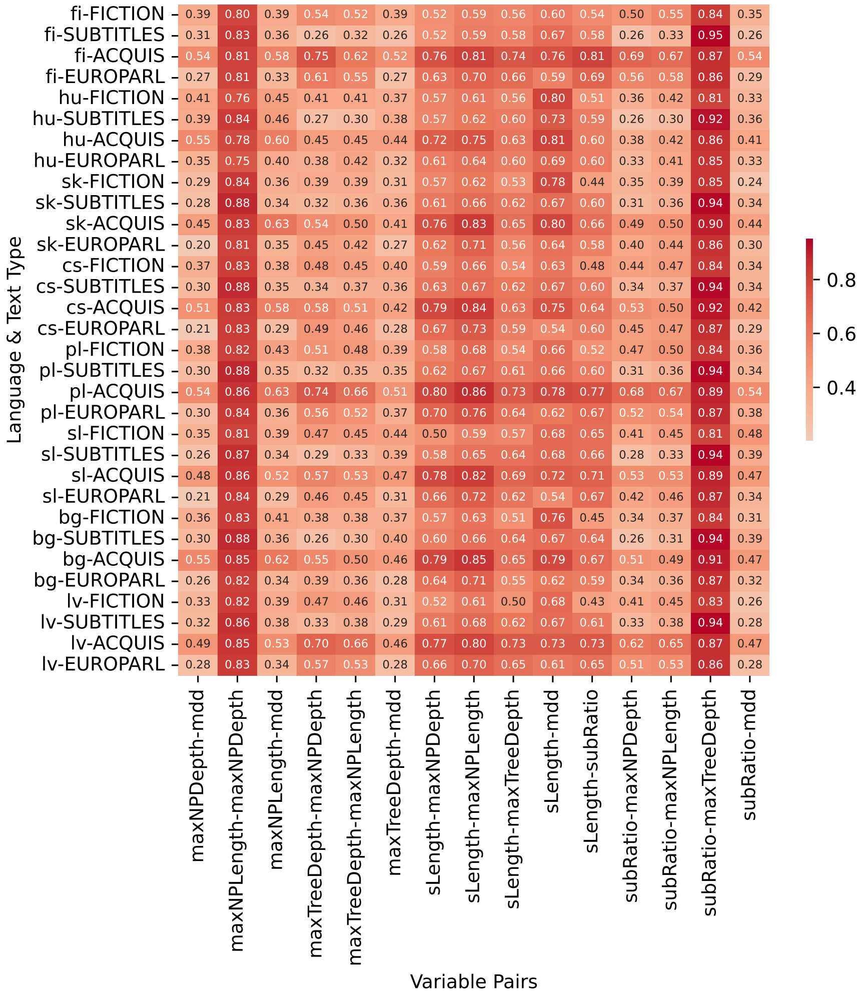

Fig. 5: Correlation patterns of syntactic complexity metrics across languages and genres (part 2).

metrics (with Subtitles leading in all 17 languages) and the two NP metrics (leading in 15 languages). In these pairs, Acquis consistently ranks second to Subtitles, indicating a strong correlation between depth and width (length) of sentences and NPs in both genres, despite the actual metric values (both clausal and NPs) lying at opposite ends (https://wiki.korpus.cz/doku.php/en:cnk:intercorp:verze16ud#detailed_statistics). Subtitles has the lowest values, with minimally elaborated NPs and clausal hierarchies reflecting its proximity to spoken registers (Levshina 2017a), whereas Acquis reaches the highest values due to its informational nature and elaborated sentence and NP structure.

The two remaining genres, Fiction and Europarl, display less pronounced tendencies, but Fiction exhibits greater heterogeneity: in 92 language–metric pair combinations (36.09% of cases), it has the lowest correlations. This heterogeneity may be attributed to the wide variation of syntactic complexity in fictional texts, including both dialogue and narrative sequences (Ebeling and Ebeling 2020). Moreover, fiction sometimes pushes the structural potential of language to its limits for stylistic purposes (see Biber and Gray 2016: 132).[8] Consequently, crosslinguistic studies based on fiction should explicitly account for style as an additional factor.

4.2 Differential analysis of metrics

In this section, we explore the language–genre interplay using random forest classification models, allowing us to assess whether the models classify languages or genres more accurately and to identify which metrics best distinguish between languages and which between genres.[9]

4.2.1 Genre and language attribution models

When it comes to genre attribution, the results show that the model achieves consistently high accuracy across all languages (Figure 6) and high F1 scores for all genres across languages (Figure 7).

Figure 7 shows that the F1 scores, indicating the degree of accuracy of genre attribution, are most even in Subtitles and Acquis, whereas Fiction and Europarl exhibit greater variation. However, even the lowest score is very high (0.81 for Latvian Europarl),

8 In our subcorpus, the variability of fiction may be amplified by the stylistic imprint of source texts, given that the various sections of the subcorpus include both original and translated texts (see Section 3).

9 For some languages and text types the sample sizes may be rather small. The number of texts in the samples is determined by the size of the smallest compared group. The tests are performed on 30% held-out parts of the sample.

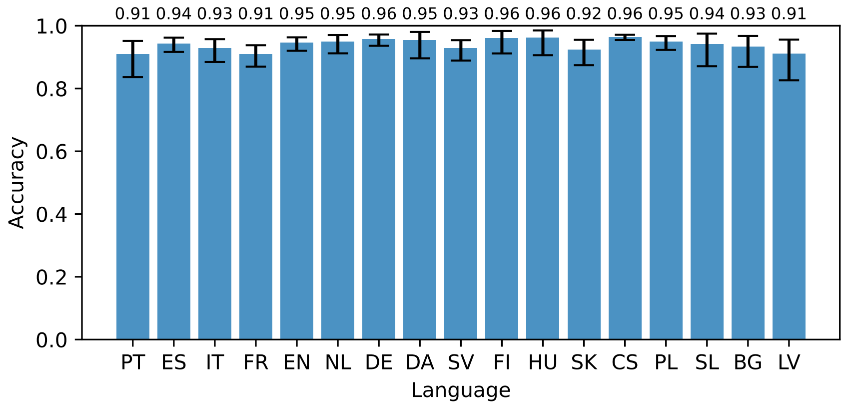

Fig. 6: Accuracy comparison of genre classification across languages (with 95% confidence intervals).

suggesting that SCMs are strong indicators for genre classification. Nonetheless, as shown in Section 4.2.3, individual metrics vary in their discriminative power across genres (see similar observations in Brunato et al. 2020: 7149).

Language identification (Fig. 8) yields much lower accuracy than genre attribution (Fig. 6). This is unsurprising, given the higher number of target categories (17 languages versus four genres). However, a closer analysis reveals important insights into the interaction between language and genre in relation to SCMs.

As shown in Figure 8, the lower accuracy of language identification is particularly evident in Fiction and Europarl. This can be attributed mainly to the high internal variation in syntactic complexity within both genres (see a similar observation for fiction in Biber and Gray 2016).[10]

Figure 9 shows that in some genres, some languages were practically indistinguishable, particularly some Slavic languages in Europarl (Slovak, Czech, and Slovenian) and Spanish in Fiction. By contrast, Acquis and Subtitles show consistently high F1 scores, with the main exception of frequent misclassification between the closely related Czech and Slovak. A deeper understanding of classification errors requires analysis of confusion matrices that display the misclassified pairs.

10 In Fiction, another factor potentially influencing the results is the small size of the training dataset: a total of 1,105 texts, with 65 per language and 46 (70%) used for training. The internal variation of Europarl is likely due to the very large number of different authors (speakers), each contributing short texts – an average of 205 words. The coherence observed in Subtitles may be reinforced by the consistent constraint on the length of the text window (see also Levshina 2017a).

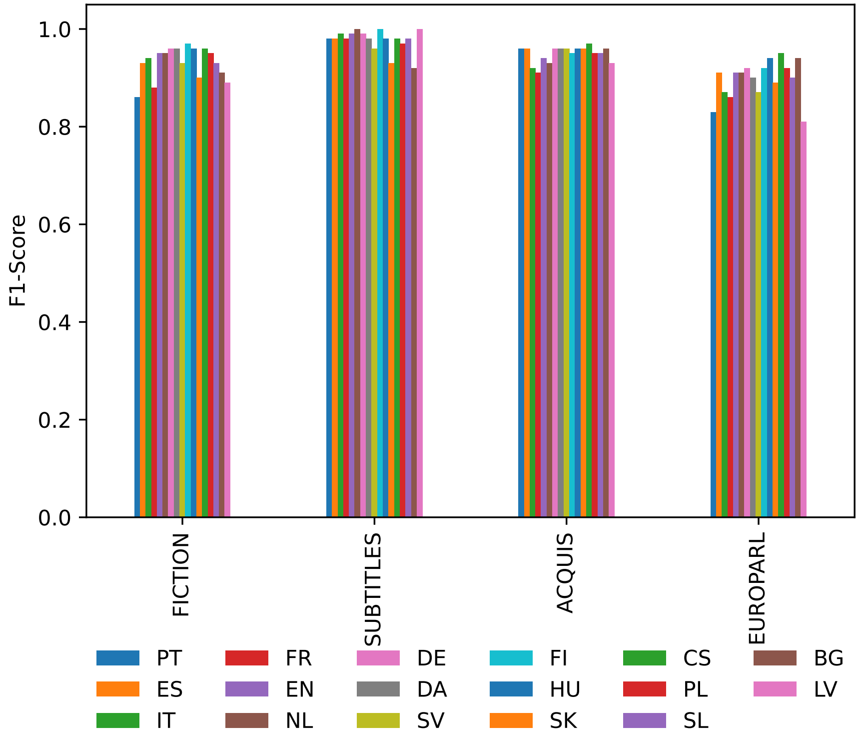

Fig. 7: F1 score comparison of genre classification across languages.

4.2.2 Confusion matrices for genre and language classification

We begin by presenting confusion matrices for genre classification by language (Figure 10), which reveal few consistent misclassification patterns across languages, as genres are rarely confused (see also Figures 6 and 7). Most observed misclassifications involve Fiction, typically confused with Acquis or Europarl.

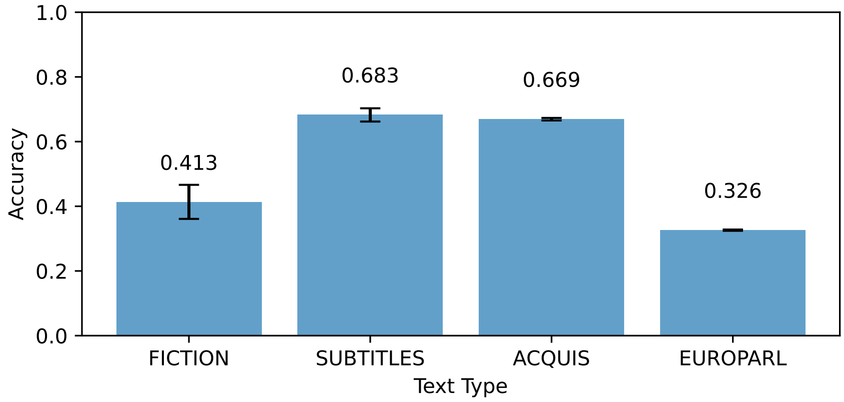

Fig. 8: Accuracy comparison of language classification across genres (with 95% confidence intervals).

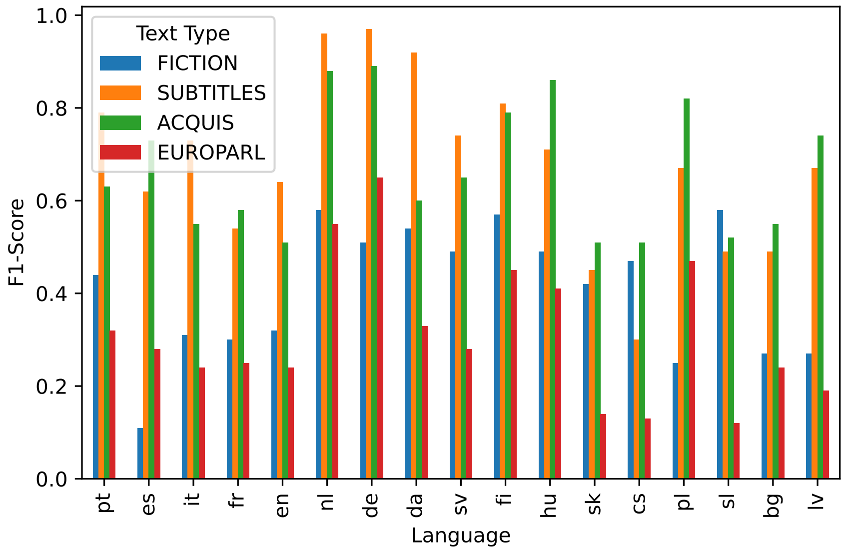

Fig. 9: F1 score comparison of language classification across genres.

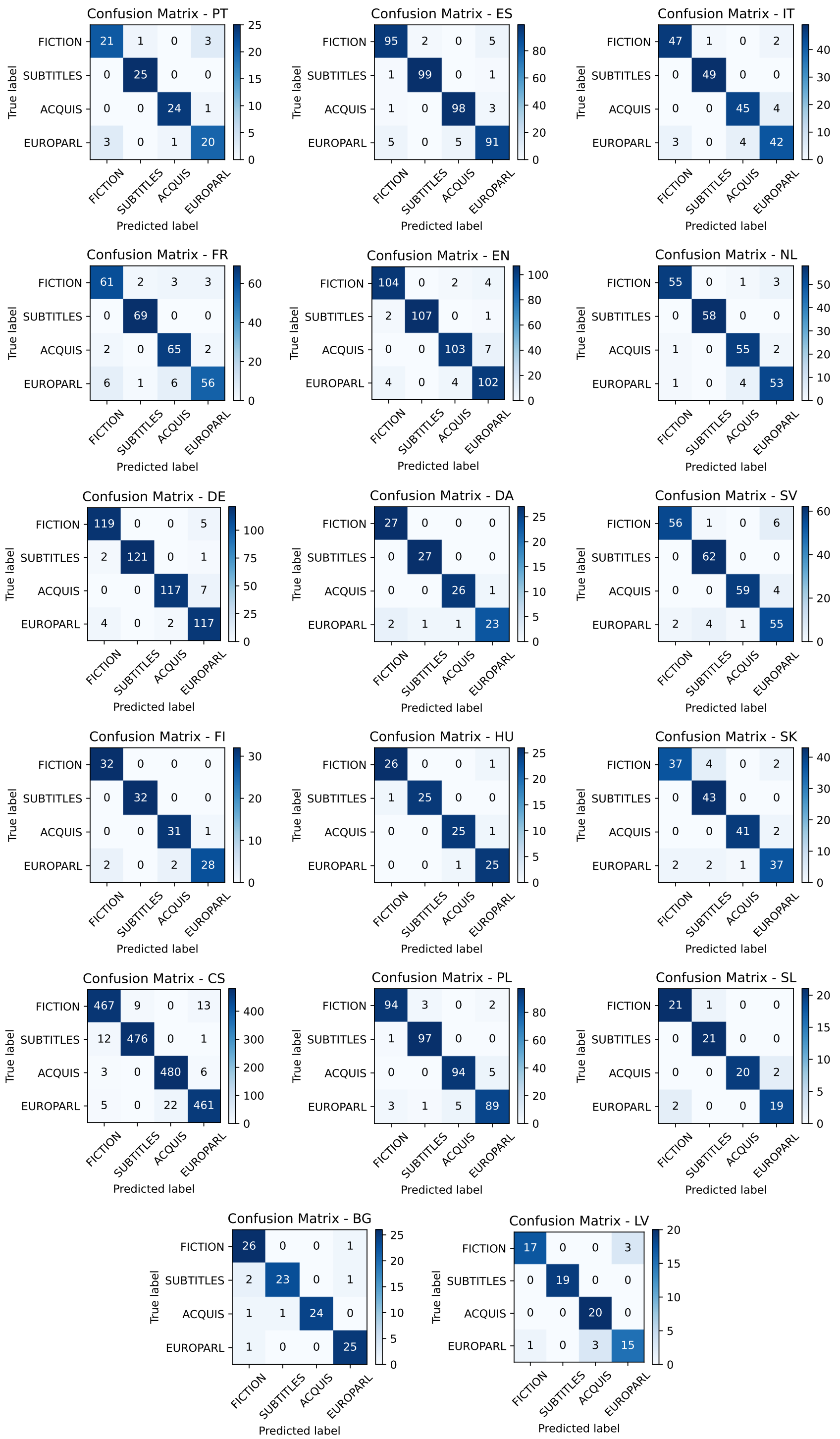

Fig. 10: Confusion matrices for genre classification across all languages.

In contrast, confusion matrices for language classification by genre, presented in Figures 11–14, reveal more nuanced patterns. Some of the misclassifications align with language genera, likely due to structural similarities. A Romance cluster is clearly visible (though less so in Subtitles) and there is a weaker Slavic cluster (Bulgarian, Czech, Polish, Slovak, Slovenian). As noted earlier, Czech and Slovak are frequently confused, given their close similarity (see especially Figure 12).

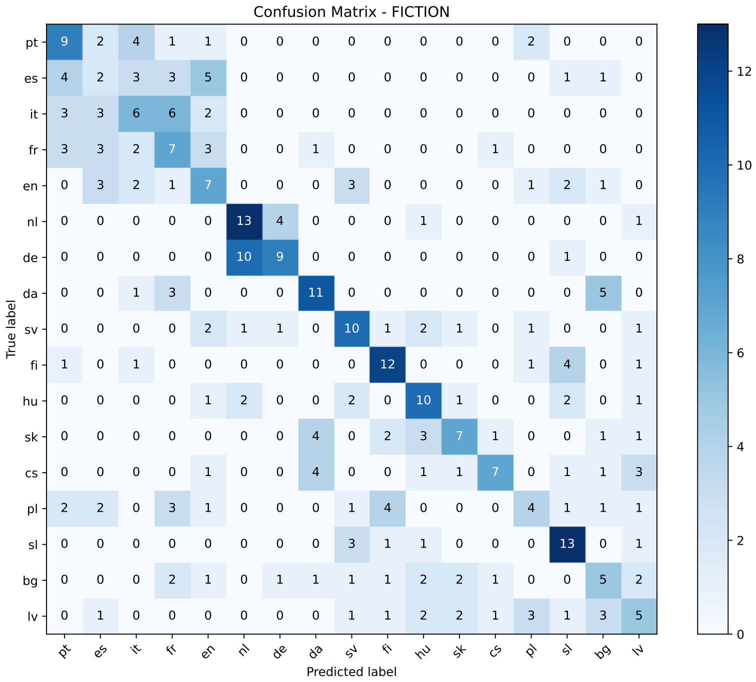

Fig. 11: Confusion matrix for language classification in Fiction.

Within Germanic, Dutch is often misclassified as German, particularly in Fiction and Europarl. Interestingly, English, however, is more often misclassified as a Romance language – especially as French in Subtitles and Europarl – than as another Germanic language, suggesting closer similarity to the Romance group regarding the SCMs (cf. a similar finding in Sadeniemi et al. [2008: 204], although in that case based on word

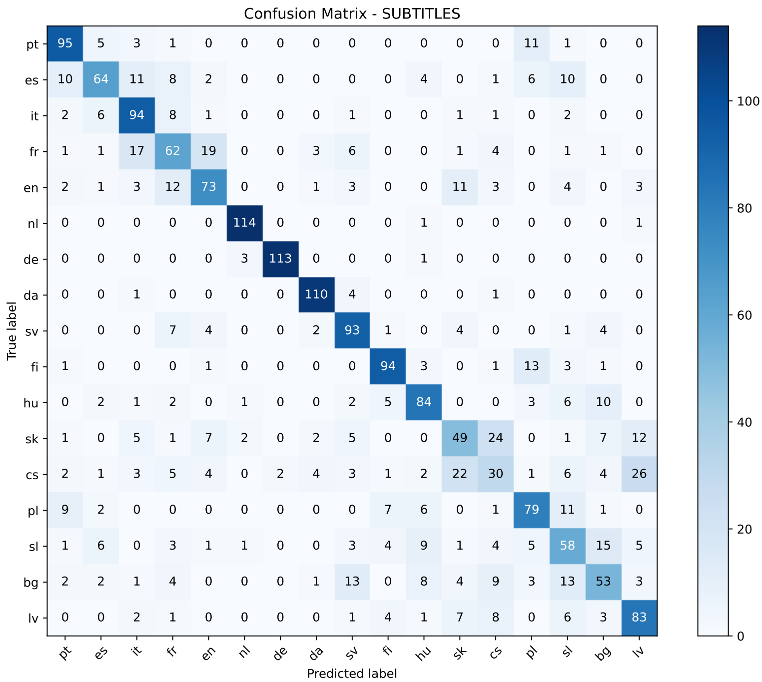

Fig. 12: Confusion matrix for language classification in Subtitles.

properties).[11] Some other patterns also cut across genera – for example, Latvian and Finnish are sometimes confused in Acquis and Europarl. These findings call for further investigations into the specific linguistic features triggering such misclassifications.

4.2.3 Feature importance analysis

A key part of this research is feature importance analysis. For genre classification, the mean values of maxNPDepth and maxNPLength ranked highest, confirming the strong relevance of NP metrics for (intra-lingual) genre variation, as already noted in Section 4.1. Mean values of sLength and maxTreeDepth also scored highly, followed by the standard deviation of sLength. In contrast, mdd and subRatio (both in terms of mean

11 English is also confused with Spanish in Fiction; e.g., Vargas Llosa's *Conversación en la catedral* and the Spanish translation of Coelho's *O Alquimista* were classified as English texts.

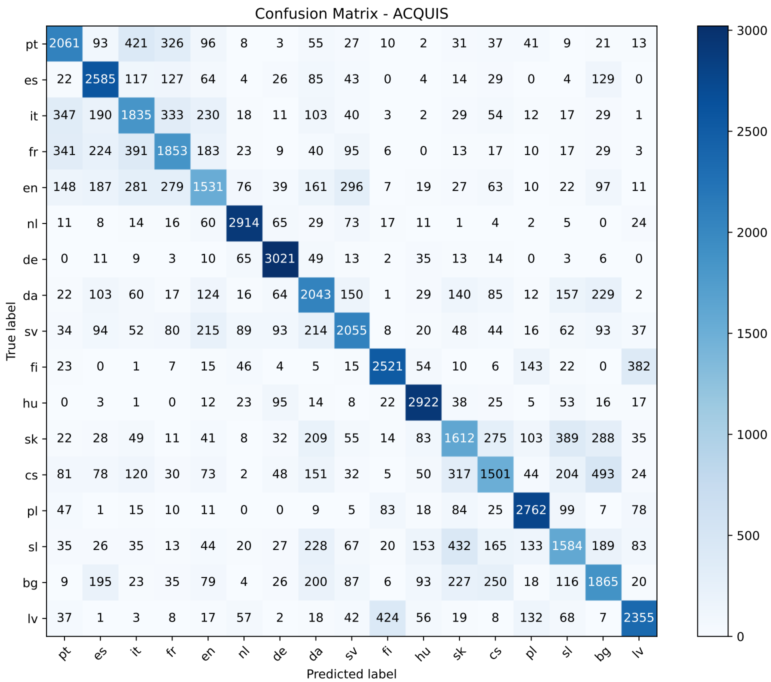

Fig. 13: Confusion matrix for language classification in Acquis.

and standard deviation) had only limited impact in genre classification. These patterns were relatively consistent across all languages, as detailed in Figure 15.

A different pattern emerged for language classification. Figure 16 shows that mdd and its standard deviation are the strongest metrics, indicating crosslinguistic reliability, possibly due to the relatively weak correlations of mdd with the remaining metrics. Other metrics showed balanced importance; even subRatio, though unhelpful for genre classification, played a role in language identification. Further crosslinguistic research should test whether differences in syntactic complexity relate to structural features such as the use of converbs and other nonfinite verb forms, which previous studies identified as an important source of complexity variation (cf. Section 2).

Overall, as detailed in Figure 16, for language classification, both average metric values and their dispersion within texts (i.e., standard deviation) were relevant. This suggests that languages differ not only in the mean values of SCMs, but also in their distributions, which highlights new directions for further research.

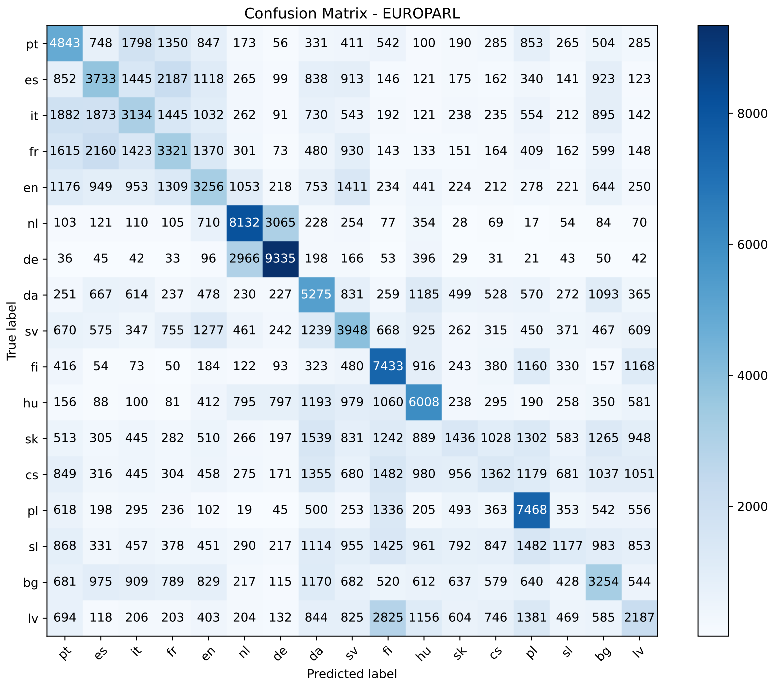

Fig. 14: Confusion matrix for language classification in Europarl.

5 Conclusions

This study investigated how crosslinguistic and genre differences influence the behavior of SCMs, focusing on their inter-correlations and effectiveness in language and genre classification. Based on a subcorpus of 17 languages and four genres from InterCorp (release 16ud), we observed substantial variation across both dimensions – in correlation patterns, classification accuracy, and feature importance.

The correlation analysis revealed stable patterns across languages and genres, notably strong correlations between clausal metrics (subRatio and maxTreeDepth) and between NP metrics (maxNPDepth and maxNPLength). In contrast, mdd showed the weakest correlations overall (except with sLength), suggesting it is the most distinctive metric. However, some patterns were genre-specific, revealing intra-linguistic variation in SCM behavior. Acquis and Subtitles showed the highest correlations across metrics and languages, indicating homogeneity of these genres. Fiction and Europarl exhibited the lowest correlations, likely due to the high number of different authors in Europarl

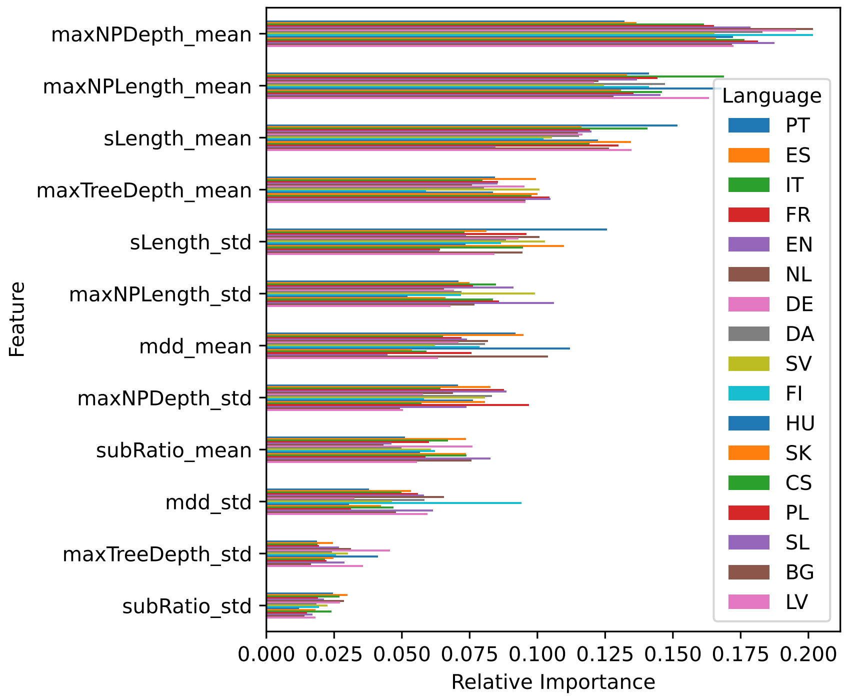

Fig. 15: Feature importance comparison for genre classification.

and to the high stylistic variability in Fiction. These findings confirm the crucial role of genre in syntactic complexity and its relevance for crosslinguistic analysis.

In classification tasks, SCMs were strong indicators of genre (cf. Cimino et al. 2017), but less effective for language – especially in Fiction and Europarl, likely due to internal variability of these genres. Feature importance analysis underscored this divide: NP metrics were most salient for genre classification, consistent with Biber and Gray (2016), while mdd – largely irrelevant for genre – emerged as decisive for language classification. Both its mean and standard deviation were relevant, suggesting that dispersion, not just central tendency, is a meaningful factor in SCM behavior.

Our future research on syntactic complexity should include a broader range of non-Indo-European languages to address the Indo-European bias of this study and further explore the potential of the InterCorp multilingual corpus. It should also examine differences in the SCM values and their underlying linguistic features, ideally through detailed analyses of parallel concordances. Crucially, both genre and language must be treated as key factors in such analyses.

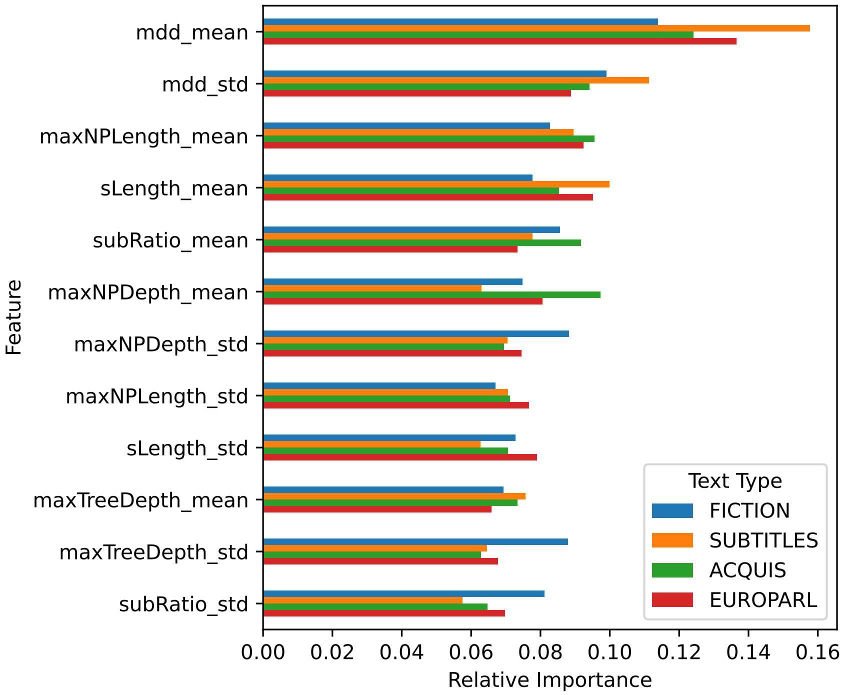

Fig. 16: Feature importance comparison for language classification.

Data availability statement

All data, scripts, and results are available on OSF at https://doi.org/10.17605/OSF.IO/KWXVS including a lightweighted version, both under the Creative Commons Attribution 4.0 International License (CC BY 4.0), which is the most permissive license that allows for maximum reuse while requiring attribution.

AI disclosure statement

The creation of scripts as well as the protocol and some parts of the article were consulted with Claude 3.7 Sonnet and GPT-4o. However, all outputs were thoroughly reviewed and edited by the authors, who bear full responsibility for their accuracy.

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